HOJUNE KIM

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EDUCATION

Stanford University *CA, United States*
M.S./Ph.D. in Aeronautics and Astronautics 09/2025 – Present
- GPA: 4.17/4.3, Advisor: Mac Schwager, Ph.D.

Seoul National University (SNU) *Seoul, South Korea*
B.S. in Aerospace Engineering, summa cum laude 03/2019 – 08/2025
- GPA: 4.21/4.3(Major), 4.07/4.3(Overall) * 1.5-year absence for mandatory military service

PUBLICATIONS

1. **H. Kim**, T. Chen, J. Sun, L. W. Osterberg, Q. Chen, K. Wang, M. Schwager. LEGS: Fine-Tuning Teleop-Free VLAs for Humanoid Loco-manipulation in an Embodied Gaussian Splatting World. *CoRL 2026*.
2. **H. Kim**, H. Jang, A. Kim. 2D Ego-Motion with Yaw Estimation using Only mmWave Radars via Two-Way Weighted ICP. *IEEE ICRA 2024 Workshop on Radar in Robotics*, Yokohama, 2024.
3. **H. Kim***, J. Lee*, G. Mesesan, R. Giubilato, R. Schuller, K. Fründ, J. Lee. Knowledge-enabled Adaptive Locomotion Using Semantic SLAM with Active Learning. *In preparation*.

* Equal contribution

RESEARCH EXPERIENCE

Stanford University – Multi-robot Systems Lab *CA, United States*
Research Assistant, Advisor: Mac Schwager, Ph.D. 09/2025 – Present
- Led LEGS, a simulator that composites physics-simulated meshes over photorealistic Gaussian Splatting scenes to generate teleop-free demonstrations for zero-shot sim-to-real transfer
- Fine-tuned 3 VLA backbones ($\pi_{0.5}$, GR00T N1.6, Ψ_0) on synthetic data only, matching or exceeding teleop-trained policies on a Unitree G1 at over 15$\times$ lower cost per new scene
- Developing an agentic real-to-sim pipeline that refines a photorealistic digital twin through system identification from robot executions, verifying motions in simulation before execution

German Aerospace Center (DLR) – Humanoid TORO Team *Munich, Germany*
Guest Researcher, Advisor: Jinoh Lee, Ph.D. 09/2024 – 02/2025
- Built an adaptive locomotion system using semantic SLAM and knowledge-based reasoning to tune 8 gait parameters of a DCM-based controller, contributing to the team's 1st place at the IEEE-RAS Humanoids Competition
- Developed a scalable semantic segmentation model on Jetson Orin AGX by fusing a stream-based active learning classifier with SAM, trained through a HoloLens human-robot interaction loop

ETH Zurich *Zurich, Switzerland*
Robotic Systems Lab*, Research Assistant*, Advisor: Cesar Cadena, Ph.D. 02/2024 – 06/2024
- Trained ANYmal crowd-navigation RL policies via teacher-student distillation (PPO in Isaac Lab, GRU memory), achieving a 74.5% success rate in static and dynamic environments

Vision for Robotics Lab*, Research Assistant*, Advisor: Margarita Chli, Ph.D. 02/2024 – 08/2024
- Developed a symbolic Gaussian Belief Propagation (GBP) solver in C++ for continuous-time SLAM, achieving a 2.4$\times$ to 110.3$\times$ speedup over conventional methods
- Benchmarked distributed GBP against the centralized NLLS solver (Ceres) for continuous- and discrete-time SLAM on hand-collected data

Seoul National University – Robust Perception and Mobile Robotics Lab *Seoul, South Korea*
Research Assistant, Advisor: Ayoung Kim, Ph.D. 02/2023 – 01/2024
- Proposed a 2D odometry method using only mmWave radar, reducing RPE by 85–90% on indoor datasets with a bidirectional weighted ICP algorithm
- Designed a handheld multi-sensor system and performed camera-LiDAR-radar joint calibration via graph optimization

Industry Experience

Foundation Robotics *CA, United States*
AI Engineering Intern 06/2026 – 09/2026

- Led development of a photorealistic Gaussian Splatting simulator to verify robot URDFs and generate synthetic loco-manipulation data for teleop-free VLA training
- Built a closed-loop evaluation pipeline for testing policies in simulation before real-robot deployment

Dentronic *Palo Alto, United States*
Co-Founder & CTO 07/2024 – 02/2026

- Built a back-drivable robotic assistant arm that automates suction and retraction for solo dental procedures, deploying 50 units to paying customers across the US and Korea and raising a $1M seed round

Invited Talks

Amazon Personal Robotics Group (PRG): *Photorealism at Scale: Breaking the Humanoids Data Bottleneck* 09/2026
Bay K-Robotics: *Photorealism at Scale: Breaking the Humanoids Data Bottleneck* 08/2026

Awards

SNU Alumni Association Chairman's Award (sole recipient, College of Engineering) 08/2025
1st Place, IEEE-RAS Humanoids Competition, Adult-sized Biped Free Walk (8 teams) 11/2024
Minister of National Defense Award, Korea Defense Startup Challenge (out of 953 teams) 12/2022
Gold Prize, International Student Car Competition, Autonomous Driving (out of 60 teams) 10/2021

Academic Service

Reviewing

· Journals: IEEE Transactions on Robotics (T-RO), 2026
· Conferences: IEEE-RAS Humanoids, 2026

Teaching

· Teaching Assistant, Basics of Robot Programming and Mechanical System Design, SNU (Spring 2023)

Scholarships

Chungsoo Overseas Scholarship | Chungsoo Scholarship Foundation 2025 – 2027
Korea-Germany Junior Research Fellowship | Max Planck POSTECH/KOREA 09/2024
Global Leadership Program Scholarship | SNU 02/2024
Kwanjeong Undergraduate Scholarship | Kwanjeong Educational Foundation 2021 – 2024

Leadership & Activities

Fire Butterfly, Autonomous Drone Association 02/2023 – 08/2023
Lead Software Engineer

- Engineered a drone auto-landing system with sub-5 cm accuracy using C2 Bézier-curve planning, validated in Gazebo and PX4 on-board flight and transferred to a private company

Army Aviation Operations Command, Republic of Korea Army 08/2021 – 02/2023
CH-47D Helicopter Crew Member, Sergeant

- Maintained CH-47D helicopters and flew tactical missions, including over 60 flight hours on two volunteered wildfire suppression missions (2022 Uljin, 2022 Yeongdeok)

AI Tech Play, Non-Profit Organization for AI Education 02/2021 – 08/2021
Co-organizer & Hardware Leader

- Hosted an AI camp and RC-car competition for over 200 children nationwide with MIT Lincoln Laboratory; led the team that built the ROS2-controlled RC car and 7 Python tutorials

SNU ZERO, Autonomous Driving Car Association 01/2021 – 08/2021
SLAM Engineer

- Built EKF-based GNSS-inertial navigation with line-detection correction and camera-LiDAR fusion for obstacle avoidance on a vehicle designed from the chassis up

Patents

Parking Location Tracking System, *KR102291377B1*, 2021 (**H. Kim**, T. Kim, J. Na, J. Lee, S. Jeong)
Document Editing System Supporting Online Document Creation with Search Function, *KR100143446*, 2022 (**H. Kim**, J. Choi)